

APPLICATIONS SECTION
MULTIVARIATE TESTS
OF THE CAPITAL ASSET PRICING MODEL:
THE SOUTH AFRICAN EVIDENCE

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Summary: A recent development in financial research has been the emergence of a multivariate approach for testing asset pricing models. In this article a multivariate test for the validity of the Capital Asset Pricing Model (CAPM) on the Johannesburg Stock Exchange (JSE) is proposed. A technique for testing additional factors for asset pricing is also considered. The results show that the validity of the CAPM cannot be rejected on the JSE. By contrast to results found on the New York Stock Exchange, none of the additional factors, namely, firm size, dividend yield and liquidity were found to influence asset pricing on the JSE.

1. INTRODUCTION

Economists have long been interested in determining the factors that affect the pricing of financial assets such as shares and bonds. In the 1950s Markowitz (1952, 1959) developed a theory of choice, that is, of portfolio selection amongst financial assets, but said nothing about the pricing of these assets in financial markets. In the mid 1960s Sharpe (1964), Lintner (1965)

and Mossin (1966) developed a theory on the *pricing* of financial assets which culminated in the Capital Asset Pricing Model (CAPM). This model has subsequently played a central role in the finance literature. Modern investment techniques have been strongly influenced by the CAPM and its provocative implications. It is understandable, therefore, that there has been much interest in tests of the validity of the CAPM.

Most empirical tests of the CAPM are univariate procedures [Black, Jensen and Scholes (1972), Fama and Macbeth (1973)]. Recently, however, a multivariate test has been proposed by Gibbons, Ross and Shanken (1989). The focus of this article is to test the validity of the CAPM for listed shares on the Johannesburg Stock Exchange (JSE) using this multivariate procedure. The possible inclusion of additional pricing factors, namely dividend yield, firm size and liquidity will also be investigated.

In section 2 a brief review of Capital Market Theory, which encompasses the CAPM, is outlined. In section 3 the multivariate framework for testing the CAPM is discussed. Section 4 describes the test methodology and the data. In section 5 the results are presented, and section 6 contains some concluding remarks.

2. REVIEW OF CAPITAL MARKET THEORY

The modern theory of finance has grown with the development of the computer and with the quality of financial data, in particular stock market price series. Thus the modern theory of finance has been rooted in empirical analysis, with the major testing ground being the New York Stock Exchange (NYSE). By contrast, little empirical research has been conducted on smaller exchanges, such as the Johannesburg Stock Exchange.

2.1 Decision framework

The major challenge facing investors has always been the maximization of wealth in a world of uncertainty. The problem of choice under uncertainty is characterised by the situation where an investor faces a set of investment alternatives, and the outcomes associated with these alternatives are

uncertain. Two main conceptual frameworks have been developed to deal with the problem; the state-preference framework developed by Arrow (1951) and later by Debreu (1959), and the mean-variance or parameter-preference framework developed by Markowitz (1952). The state-preference framework, while useful for investigating theoretical issues, lacks empirical content due to the difficulty in quantifying the pay-offs offered in different states of nature. The work of Markowitz (1952, 1959), however, was based on the assumption that share returns (i.e. percentage price changes) are normally distributed, and that consequently the mean and the variance are the only statistics relevant for describing the distribution of share returns.

Markowitz (1952) showed that the portfolio selection problem is essentially a search amongst an opportunity set of shares to determine investment weights (i.e. portfolios) yielding the maximum return for given levels of variability (risk). He was able to formulate these ideas as a quadratic programming problem, and showed that the solution to this problem resulted in a set of mean-variance "efficient" portfolios, that is, efficient in the sense that for the period under consideration an efficient portfolio provides maximum expected return for a given level of risk, and minimum risk for a given level of expected return. The set of "efficient" portfolios is commonly called the efficient frontier. A consequence of the theory is that risk-averse investors would attempt to select efficient portfolios to maximize their expected utility.

The pioneering work of Markowitz laid the foundation for much research into models in the mean-variance framework. In particular, Tobin (1958) extended Markowitz' theory by introducing a risk-free asset (usually a fixed income investment). He showed that the investment decision could then be separated into two phases; first, the selection of a unique risky efficient portfolio, and second a separate choice concerning the allocation of funds between the unique portfolio and the single risk-free asset. In addition, Sharpe (1963) proposed a model (later called the market model) which reduced the number of input variables required by the Markowitz procedure

by linking each asset in the opportunity set to a common market index. These developments formed the basis of the subsequent Capital Market Theory and proved crucial in the development of the Capital Asset Pricing Model.

2.2 The notion of risk

In the Markowitz mean-variance framework, risk is measured by the variability, or more specifically the standard deviation, of returns. Thus, the risk of a portfolio can be expressed as

$$\sigma_p = \left[\text{Var} \left[\sum_{i=1}^N X_i r_i \right] \right]^{\frac{1}{2}}$$

where X_i is the proportion of funds invested in asset i ($i = 1, 2, \dots, N$), and r_i is the return on asset i . This can be rewritten as

$$\sigma_p = \left[\sum_{i=1}^N X_i^2 \sigma_i^2 + \sum_{i=1}^N \sum_{j=1, j \neq i}^N X_i X_j \sigma_{ij} \right]^{\frac{1}{2}}$$

where σ_i^2 is the variance of r_i , and σ_{ij} is the covariance of the returns on assets i and j .

To determine the relevant measure of risk for an individual asset we need to determine the impact that a change in the proportion of funds invested in that asset will have on the risk of the portfolio. Thus, we have (see for example, Elton and Gruber (1984, p. 284))

$$\frac{d\sigma_p}{dX_i} = \frac{\frac{1}{2} \left[2X_i \sigma_i^2 + 2 \sum_{j=1, j \neq i}^N X_j \sigma_{ij} \right]}{\left[\sum_{i=1}^N X_i^2 \sigma_i^2 + \sum_{i=1}^N \sum_{j=1, j \neq i}^N X_i X_j \sigma_{ij} \right]^{\frac{1}{2}}} = \frac{\sigma_{ip}}{\sigma_p}$$

where σ_{ip} is the covariance of the return on asset i and the return on portfolio p . This shows that the relevant risk of a security in a portfolio context is the covariance of the asset returns with the portfolio rather than the standard deviation (risk) of the security per se. It is this concept of risk which is central to the development of the CAPM.

2.3 The market model

In spite of the growing popularity of models in the mean-variance framework, most researchers such as Sharpe (1963) and Baumol (1963), had developed models dealing with asset choice under conditions of risk. No one had as yet attempted to extend these models to construct a market equilibrium theory of asset pricing under conditions of risk in the mean-variance framework. Sharpe (1964), Lintner (1965) and Mossin (1966) made the first major contributions in developing a theory. This theory has become known as Capital Market Theory and as a result the model is frequently referred to as the Sharpe-Lintner-Mossin Capital Asset Pricing Model, or more simply as the CAPM.

In deriving an equilibrium theory of asset pricing numerous assumptions about the nature of the market must be made (see for example, Copeland and Weston (1984), p. 186). These include assumptions that trading is frictionless (i.e. no taxes or transaction costs) and that assets are perfectly divisible (i.e. can be divided to accommodate the demand for any proportion of the asset). In addition, it is assumed that all investors are risk averse and have homogeneous expectations of asset returns (i.e. percentage price changes). Under such conditions it follows from Tobin (1958) that all investors will hold the same portfolio of risky assets, and differences in risk aversion will be accommodated merely by fluctuating the proportion of funds invested in the risk-free asset (with the remainder being invested in the portfolio of risky assets). Consequently, since in equilibrium total demand for each asset must equal total supply, the risky portfolio held by all investors must be the market portfolio – that is, the portfolio of all assets in the economy in proportion to their market capitalization.

From the development of the risk concept in section 2.2 it follows that, in equilibrium, since all investors will hold the market portfolio, the relevant risk of any asset i is proportional to the covariance between that asset's returns and those of the market portfolio (m). Standardizing this covariance by the variance of the market returns, σ_m^2 (which is the same for all assets),

yields the following measure of risk for asset i :

$$\beta_i = \sigma_{im} / \sigma_m^2$$

Elementary statistics show that this is the slope parameter in the simple linear regression

$$(1) \quad r_{it} = \alpha_i + \beta_i r_{mt} + e_{it} \quad \text{for } i = 1, 2, \dots, N; \quad t = 1, 2, \dots, T$$

where r_{it} is the return on asset i in period t ;

r_{mt} is the return on the market index in period t ;

e_{it} is the disturbance term for asset i in period t ; and

α_i and β_i are parameters unique to asset i .

The assumptions for the error term are

$$\begin{aligned} E(e_t) &= 0 \\ E(e_t, e_s) &= \Sigma & \text{for } t = s \\ &= 0 & \text{for } t \neq s \end{aligned}$$

where Σ is the $(N \times N)$ disturbance covariance matrix.

This model is known as the market model or the security characteristic line, and is used to estimate the relevant risk, β_i , for an asset. This measure of risk indicates the volatility of the asset's return relative to the market. Assets having β greater than one are regarded as being more volatile and hence more risky than the market, while securities having β less than one are regarded as being less risky than the market.

The importance of the beta coefficient is also emphasized by consideration of the variability of a security's returns using the market model. More specifically the market model can be used to show that the "total risk" of share i stems from two sources, i.e.

$$\begin{aligned} \text{var}(r_{it}) &= \text{var}(\alpha_i + \beta_i(r_{mt}) + e_{it}) \\ &= \beta_i^2 \text{var}(r_{mt}) + \text{var}(e_{it}) \end{aligned}$$

since α_i and β_i are constant. The above can be interpreted as

Total Risk of share i = Market Risk of share i + Unique Risk of share i .

The market risk of a share stems from the fact that there are economy-wide factors which affect all securities. These include factors such as movements in interest rates, the business cycle, or the inflation rate, which affect almost all firms to a degree. On the other hand, the unique risk is the risk embodied in individual companies, for example local strikes or bad management.

These ideas can be extended to consider the risk of portfolios as well as individual shares. In particular, it is easy to show that for a well-diversified portfolio, market risk is the only risk that is relevant since unique risk will be diversified.

It is evident, therefore, that the contribution a security makes to the risk of a well-diversified portfolio depends only on how sensitive it is to movements of the market as a whole. This sensitivity, in essence, is measured by the beta coefficient of the security. Hence the beta coefficient has come to play an important role in the modern finance literature as a measure of market risk.

2.4 The Capital Asset Pricing Model

The CAPM, in essence, embodies the notion that investors require sufficient reward for their delayed consumption (i.e. to maintain the purchasing power of their investment), as well as some additional return for taking on risk. That is, the return an investor would expect on a risky asset i (i.e. the discount rate), is

$$E(r_i) = r_f + \pi_i$$

where $E(r_i)$ is the expected return on asset i ; r_f is the risk-free rate of interest; and π is the premium for bearing risk.

In section 2.3 it was argued that risk-averse investors would diversify all firm-specific risk, and consequently the CAPM states that investors can only expect to be compensated for bearing non-diversifiable, or market risk (i.e. β).

Clearly, an investor can construct a portfolio with any given beta and no

unique risk by investing a proportion, β , in the market portfolio and the remainder, $1 - \beta$, in the risk-free asset. This portfolio has expected return

$$E(r) = \beta E(r_m) + (1 - \beta)r_f$$

Thus under conditions of market equilibrium, any share with beta coefficient β_i equal to β must offer the same return to be attractive (and cannot offer more as there would be excess demand). Hence the above equation must hold for every share under conditions of market equilibrium. This has become known as the CAPM and the terms can be rearranged to give the familiar form

$$(2) \quad E(r) = r_f + \beta[E(r_m) - r_f]$$

A formal proof can be found in financial texts such as Copeland and Weston (1984). In the framework of Markowitz equation (2) implies that the market portfolio, consisting of all assets in their exact market capitalization proportions, is expected to be the optimal portfolio. This is an important point and Roll (1977) has emphasized that testing the CAPM is in fact equivalent to testing the assertion that the market portfolio is located on the frontier of efficient portfolios in mean-variance space.

The CAPM is widely accepted as a reasonable theoretical development, and consequently plays a central role in financial theory. The fundamental question is whether the model is empirically supported, and this question is the focus of this paper.

In the early 1970s the prominent studies of the validity of the CAPM by Black, Jensen and Scholes (1972), Blume and Friend (1973) and Fama and Macbeth (1973) on the NYSE used univariate testing procedures, and found general support for the CAPM. Recent empirical evidence on the JSE presented by Bradfield, Affleck-Graves and Barr (1988) using similar testing procedures reveals that the CAPM appears to be a reasonable model for JSE stocks.

It is well known that an opportunity set of shares has essentially a multivariate structure. Consequently the multivariate testing procedure

proposed by Gibbons, Ross and Shanken (1989) has recently captured the interest of several empiricists. This testing procedure will be used in the next section to re-examine the validity of the CAPM for JSE stocks.

Recent attention has also focused on whether additional explanatory factors are relevant in the CAPM. In particular, factors such as dividend yield, market capitalisation (or firm size), liquidity effects and others have been empirically tested for evidence that may promise the investor consistent excess returns over the market. Since the above-mentioned additional pricing factors are a secondary focus of this article, a brief discussion of the pertinent issues relating to these factors follows.

2.4.1 The dividend factor

The interest in the dividend yield stems from the effect of differential tax rates on capital gains and dividends. Dividends have generally been more heavily taxed than capital gains and consequently it is argued that the higher a securities dividend yield, the higher the pre-tax return an investor requires to compensate for the tax liability incurred. Counter arguments have been put forward suggesting investors are able to effectively transform dividend income into capital gains, and it has been argued that the pre-tax return on high dividend yielding securities should not differ from that of low dividend yielding securities (holding risk constant). Black and Scholes (1974) pioneered the empirical testing of the effects of dividend yield on common stock returns on the NYSE. They did not, however, find evidence of a significant dividend yield effect. Similar testing procedures employed by Bradfield, Affleck-Graves and Barr (1988) on the JSE show no significant differences between the returns on high dividend and low dividend yield stocks on the JSE.

2.4.2 The firm-size factor

Recent studies on stock markets have documented overwhelming evidence of a significant relation between common stock returns, and the market value of

common stock. This has become known as the 'size effect'. Banz (1981) was one of the first to investigate this relationship on the NYSE, and found a negative statistical association between returns and firm size. This implies that shares of firms with large market values have had smaller returns on average than small firms in similar risk classes. Similar results have been documented on the NYSE by Reinganum (1981); Blume and Stambaugh (1983) and Stoll and Whaley (1983). Surprisingly, no evidence of a 'size effect' on the JSE was found by Bradfield, Affleck-Graves and Barr (1988) as well as De Villiers *et al.* (1986).

2.4.3 The liquidity factor

Very little empirical evidence of a liquidity effect on stock returns has emerged in the literature. However, in a recent paper by Amihud and Mendelson (1986) empirical evidence on the NYSE was found which indicates that liquidity (the buy-sell spread) is significantly related to return in the CAPM framework. The implication of this finding is that investors should require a higher expected return for less liquid stocks to compensate them for the higher cost of trading. Bradfield, Affleck-Graves and Barr (1988), however, found no evidence of a liquidity effect for JSE stocks.

All findings discussed in section 2.4 were obtained using the univariate testing procedures. In this paper the existence of these factors is tested in a multivariate framework.

3. MULTIVARIATE FRAMEWORK FOR TESTING THE CAPM

The Gibbons, Ross and Shanken (1989) (henceforth GRS) test statistic is derived under the assumption of a given risk-free asset having return r_{ft} for each time period. Furthermore GRS assume that excess returns of all assets under study are independent and identically distributed as multivariate normal variates through time. The excess return for each asset can be written as:

$$R_{it} = r_{it} - r_{ft} \quad t = 1, 2, \dots, T \quad \text{for each asset } i$$

and the excess return on the market portfolio can be written as

$$R_{mt} = r_{mt} - r_{ft} \quad t = 1, 2, \dots, T$$

The problems of using historical data to test a model formulated in terms of expectations are well known, and usually unavoidable. With this in mind GRS rearrange the CAPM as follows:

$$E(r_i) - r_f = \beta_i [E(r_m) - r_f]$$

which is equivalent to:

$$(3) \quad E(R_i) = \beta_i E(R_m)$$

GRS compare (3) to the excess return form of the Market Model (1), i.e.

$$(4) \quad R_i = \alpha_i + \beta_i R_{mt} + e_{it}$$

which has the same underlying assumptions as (1).

Combining (3) with the distributional assumption given by (4) GRS point out that testing the validity of the CAPM is equivalent to testing the null hypothesis that the α_i in (4) are equal to zero, that is

$$H_0: \alpha_i = 0 \quad \text{for all } i = 1, 2, \dots, N$$

GRS assert that the joint significance of the estimated α_i can be reported by first estimating the multivariate system of (4) using ordinary least squares for each individual asset. They argue that the estimated intercepts have a multivariate normal distribution centred at the true parameter vector with variance given by

$$\text{Var}(T^{\frac{1}{2}} \hat{\alpha}) = (1 + \theta_m^2) \Sigma$$

where T is the number of time series observations used in the estimation procedure;

$$\hat{\alpha} = (\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_N)'$$

$$\theta_m = \bar{R}_m / s_m$$

$$\bar{R}_m = \frac{1}{T} \sum_{t=1}^T R_{mt}$$

where $\hat{R}_{mt}$ is defined as the estimated return on the market during time period t ; and

$$s_m^2 = \frac{1}{T} \sum_{t=1}^T (\hat{R}_{mt} - \bar{R}_m)^2$$

Furthermore under the null hypothesis:

$$T(1 + \hat{\theta}_m^2)^{-1} \hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha} \sim \chi_N^2$$

However, since Σ is not generally known, GRS show that

$$(5) \quad \Gamma_1 = [T/(T-2)][(T-N-1)/N] W \sim F(\lambda)_{N; T-N-1}$$

where

$$W = (1 + \hat{\theta}_m^2)^{-1} \hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha};$$

$\hat{\Sigma}$ is the unbiased sample covariance matrix of the residuals;

$F(\lambda)$ is the noncentral F distribution with noncentrality parameter λ ; and

$$\lambda = T(1 + \hat{\theta}_m^2)^{-1} \alpha' \Sigma^{-1} \alpha$$

Under the null hypothesis the non-centrality parameter equals zero and the distribution of Γ_1 has a central F distribution. Hence Γ_1 can be used as the test statistic for testing the null hypothesis $\alpha = 0$.

An important contribution by GRS is that they are able to give a plausible economic interpretation of their test statistic using a geometrical approach. In particular, GRS show that

$$(6) \quad W = \left[\frac{1 + \hat{\theta}^{*2}}{1 + \hat{\theta}_m^2} \right]^2 - 1 \equiv \psi^2 - 1$$

where

$$\psi = \frac{1 + \hat{\theta}^{*2}}{1 + \hat{\theta}_m^2}$$

and $\hat{\theta}^*$ is the observed maximum mean return per unit of standard deviation (this is also referred to as the Sharpe reward-to-variability ratio of the observed optimal portfolio). Recall that $\hat{\theta}_m = \bar{R}_m / S_m$ (equivalently the Sharpe reward-to-variability ratio of the market portfolio).

Clearly, in (6) ψ^2 should be close to one under the null hypothesis. When $\hat{\theta}^*$ is sufficiently greater than $\hat{\theta}_m$, the return per unit of risk for the market portfolio is much lower than that of the observed optimal portfolio,

and consequently the hypothesis that the market portfolio is expected to be optimal is likely to be rejected. (Recall that the validity of the CAPM and the expected optimality of the market portfolio form a joint hypothesis as argued by Roll (1977)).

The geometrical interpretation of the test statistic is shown in figure 1. From the figure it can be seen that ψ is the distance along the line OP up to any given risk level, σ , divided by the similar distance along the line OM. Note the distance Oa is equal to $\sqrt{1 + \hat{\theta}^{*2}}$ and the distance Ob is equal to $\sqrt{1 + \hat{\theta}_m^2}$. The further M lies from the position of the observed optimal portfolio, P, the larger the test statistic will be, thus increasing the likelihood of rejection of the null hypothesis.

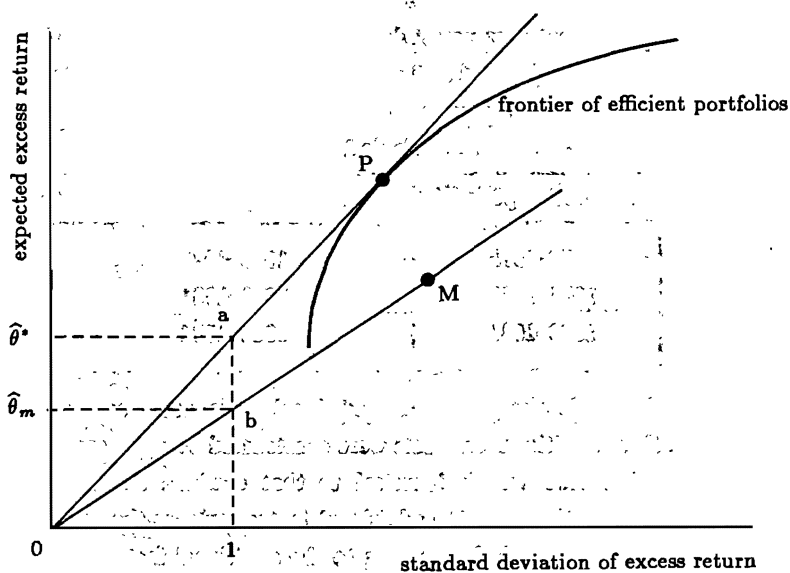


Figure 1. Geometrical interpretation of the test statistic

4. METHODOLOGY AND DATA

The data consisted of all shares listed on the JSE with a complete price record over the period from 1 January 1973 to 31 December 1984. This constituted a total of 390 listed securities. To avoid measurement errors a rather elaborate procedure of grouping shares into portfolios has traditionally been used. The approach of grouping shares into portfolios for testing the CAPM was initially suggested by Fama and Macbeth (1973) for their univariate testing procedures. The specifics of this approach are outlined below.

A series of 12 beta-sorted portfolios were constructed in the *traditional* manner from the data set as follows: Three years' data were initially used to estimate the betas of the 390 shares using model (4), which were then ranked according to their β and divided into 12 portfolios, whereafter the returns on these 12 portfolios were computed over the next three years. This procedure was repeated three times, rebalancing portfolios every three years, with the exception that four years' data were subsequently used in the estimation of the β . This resulted in 9 years' return data for the 12 beta-sorted portfolios. Table 1 shows the periods of construction of all of these data sets.

Table 1

Formation period	Return computation period
1973-1975	1976-1978
1975-1978	1979-1981
1978-1981	1982-1984

Since this article also focuses on additional explanatory components in the CAPM, three additional data sets were constructed over the same period. The construction procedure is identical to that outlined above, with the exception that an estimate of the additional factor replaces the role of beta. That is, securities are ranked according to the additional factor rather than beta. Firstly a liquidity-sorted data set was constructed using the liquidity measure proposed by Roll (1984). Market capitalization-sorted (as a measure of firm size) and dividend yield-sorted data sets were similarly

constructed. Due to the difficulty in obtaining complete records of market capitalization and dividend yields for the 390 securities, a systematic sample of 100 securities having the relevant complete records was selected for the latter two data sets.

The rationale behind using the multivariate procedure to test whether additional factors other than the asset's β are relevant for pricing of the asset is best explained using the geometrical interpretation of the test statistic.

The optimal mix of portfolios grouped according to some factor which has an additional influence on return (other than beta) is likely to yield a $\hat{\theta}^*$ value larger than that obtained using beta-sorted portfolios (that is, will lie in a more favourable position in risk-return space). Furthermore, if the influence of the additional factor on asset returns is statistically significant, then the distance between the market portfolio and the *ex post* optimal portfolio (obtained from the additional factor grouping) is likely to be significant, resulting in a significant value for the test statistic.

The JSE-Actuaries Overall index was used as a surrogate for the market portfolio (i.e. portfolio M), while the 12 month fixed-deposit rate quoted at major commercial banks (extracted from the South African Quarterly Reserve Bank Bulletins) was used as a surrogate for the risk-free rate, r_f . Returns were computed at four-week intervals. This was done to proxy monthly return data which has traditionally been used in studies of this nature.

The test statistic proposed by Gibbon, Ross and Shanken (1986) to test the efficiency of a given portfolio was computed for each of the data sets using equation (5).

5. EMPIRICAL RESULTS

5.1 The beta-sorted data set

Table 2 gives the summary statistics on the beta-sorted portfolios based on four-weekly return data, from 1976 to 1984 ($T = 117$, $N = 12$). All simple

returns are excess returns, with the 12 month fixed-deposit rate at major commercial banks used as a surrogate for the risk-free rate. The JSE-Actuaries Overall Index is portfolio M. The tabulated parameter estimates are for the regression model (4).

GRS (in a study of the efficiency of the CRSP Index (Centre for Research Security Prices)) present a similar table and find systematically negative alphas for high-beta portfolios, and positive alphas for low-beta portfolios. They comment that these findings are similar to those of Black, Jensen and Scholes (1972), who argued that the expected excess returns on high-beta assets are lower than the CAPM suggests, and that those of low-beta assets are higher than the CAPM suggests. By contrast, the evidence on the JSE presented in table 2 shows no systematic relationship between the portfolio alphas and betas, and it is interesting to note that all the alphas are positive.

Table 2. Summary statistics for beta-sorted portfolios

Portfolio number*	$\hat{\alpha}_i$	$t(\hat{\alpha}_i)$	$\hat{\beta}_i$	$t(\hat{\beta}_i)$	R_i^2
1	0,00715	1,89	0,33	5,34	0,20
2	0,01045	2,19	0,50	6,47	0,27
3	0,00427	1,16	0,52	8,60	0,39
4	0,00793	1,81	0,47	6,66	0,28
5	0,00396	1,12	0,56	9,67	0,45
6	0,00351	1,09	0,59	11,38	0,53
7	0,00397	1,12	0,70	12,21	0,56
8	0,00250	0,81	0,67	13,30	0,61
9	0,00789	1,84	0,89	12,75	0,59
10	0,00400	1,32	0,98	19,89	0,77
11	0,00652	2,04	1,11	21,43	0,80
12	0,01114	2,36	1,30	17,04	0,72

* Portfolio 1 consists of the low-beta firms while portfolio 12 consists of the high-beta firms. $t(\hat{\alpha}_i)$ and $t(\hat{\beta}_i)$ are the t-statistics of the $\hat{\alpha}_i$ and $\hat{\beta}_i$ respectively. R_i^2 is the coefficient of determination.

This point will be discussed in more detail after the discussion of the results presented in table 3.

Table 3 gives the summary statistics of the multivariate test of the CAPM on the JSE. The large p -value of the test statistic, Γ_1 , of the multivariate test that the $\alpha_i = 0$ in table 3 implies that the *ex ante* efficiency of the JSE-Actuaries Overall Index cannot be rejected. In other words, if this Index is taken as the true market portfolio, then the validity of the CAPM cannot be rejected for the South African case. This conclusion supports the findings of Bradfield, Barr and Affleck-Graves (1988).

Table 3. Summary statistics for the beta-sorted data set

Parameter	Statistic
$\hat{\theta}_m$	0,0828
$\hat{\theta}^*$	0,3727
$\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}$	0,132
Γ_1	1,155
p -value	0,325

$\hat{\theta}^*$ is the maximum possible excess sample mean return per unit of standard deviation.

It is somewhat surprising that all the $\hat{\alpha}_i$ of table 2 are positive, yet the multivariate test is still unable to reject the null hypothesis. Similar results are found in tables 6, 7 and 8 where the portfolios are constructed according to dividend yield, firm size and liquidity factors respectively.

To gain further insight as to why the multivariate test fails to reject this evidence, we note that GRS examined a data set based on size-sorted portfolios where 9 out of the 10 portfolios had positive alphas, yet the multivariate test failed to reject the null hypothesis. GRS argue that the estimators for α have the same pattern of correlation as the sample correlation matrix of the market model residuals. This is evident by

considering the distribution of the vector of estimated alphas, which is multivariate normal, namely

$$\hat{\alpha} \sim MVN\left(\alpha; \frac{1}{T}(1 + \theta_m^2)\Sigma\right).$$

GRS found that the estimation errors among the positive alphas were positively correlated.

The sample correlation matrix of the market model residuals based on the regressions summarized in table 2 are presented in table 4. The pattern of the correlations explain why the multivariate test fails to reject the null hypothesis. The main point to note is that all of the correlations, with the exception of two with portfolio 12, are positive, and fairly large. It thus appears as if the pattern of positive estimated values of α_i shown in table 2 is associated with the correlation in the estimation error of the market model.

Table 4. The sample correlation matrix of residuals for the beta-sorted portfolios

Portfolio number	1	2	3	4	5	6	7	8	9	10	11
2	,40										
3	,51	,45									
4	,46	,39	,58								
5	,45	,40	,60	,52							
6	,53	,52	,63	,63	,64						
7	,51	,50	,70	,53	,58	,67					
8	,50	,40	,66	,54	,60	,68	,73				
9	,32	,25	,36	,29	,33	,35	,43	,41			
10	,40	,30	,45	,35	,28	,34	,39	,42	,18		
11	,11	,05	,08	,03	,10	,01	,07	,09	,08	,52	
12	,16	,01	,02	-,07	,01	-,08	,00	,03	,05	,43	,68

* Portfolio 1 consists of firms with smallest betas while portfolio 12 consists of firms with the largest betas.

To understand the impact of these correlation coefficients on the univariate test results, suppose all twelve alphas were *independent*. Then, under the null hypothesis that $\alpha_i = 0$, for all i , the probability of all twelve $\hat{\alpha}_i$ being positive is only $(\frac{1}{2})^{12}$. However, if all twelve $\hat{\alpha}_i$ were perfectly positively correlated, then the probability that all the $\hat{\alpha}_i$ are positive is $\frac{1}{2}$. Similar arguments can be invoked to show that although six out of twelve $\hat{\alpha}_i$ were significant at the 5% level using individual t-tests, the strong positive correlations suggest that the univariate tests mis-state the results.

The results of the multivariate test found here are not unlike those of GRS who tested the efficiency of the CRSP Equal-Weighted Index on the NYSE over the 1931-1965 period. Their resulting test statistic and component statistics are summarized and compared to the results on the JSE in table 5.

It is interesting to note that the relative difference between $\hat{\theta}^*$ and $\hat{\theta}_m$ for the JSE is substantially larger than that found on the NYSE, but that the p -values do not reflect this to the same extent. This may suggest that the multivariate test has substantially less power on the JSE than on the NYSE.

5.2 Additional effects – Dividend yield, firm size and liquidity

Tables 6, 7 and 8 give the summary statistics for the dividend yield, firm size and liquidity-sorted data sets over the 1976-1984 period. The parameter estimates in these tables are for the regression model 4. In the construction of the data set sorted on dividend yield, estimates of dividend yield were computed as the total dividend paid during the year divided by the share price at the end of that year. Estimates of market capitalization were computed as the year-end share price multiplied by the number of shares outstanding. Finally, year-end liquidity estimates were computed using Rolls' (1984) measure on the preceding four years of data.

Inspection of the summary statistics in tables 6, 7 and 8 shows no systematic relationship between the $\hat{\alpha}_i$ and $\hat{\beta}_i$. For the size-sorted data

set summarized in table 7, however, it appears that the betas are positively related to firm size. Bradfield (1989), however, has shown that the beta estimates of shares that trade infrequently suffer from a downward bias caused by infrequent trading. Historically smaller companies are known to suffer from the infrequent trading phenomenon to a greater extent than

Table 5. Comparison of JSE and NYSE multivariate test results

	JSE	NYSE
N	12	10
T	117	420
$\hat{\theta}^*$	0,373	0,227
$\hat{\theta}_m$	0,083	0,166
W	0,131	0,023
F_1	1,155	0,958
p -value	0,325	0,476

Table 6. Summary statistics for portfolios sorted on dividend yield

Portfolio number*	$\hat{\alpha}_i$	$t(\hat{\alpha}_i)$	$\hat{\beta}_i$	$t(\hat{\beta}_i)$	R_i^2
1	0,0153	2,324	1,025	9,616	0,446
2	0,0155	1,866	0,981	7,282	0,316
3	0,0055	1,415	0,764	12,122	0,561
4	0,0041	1,000	0,679	10,170	0,474
5	0,0129	0,067	0,867	3,993	0,122
6	0,0078	1,712	0,793	10,776	0,502
7	0,0119	2,882	0,457	6,821	0,288
8	0,0023	0,623	1,128	19,020	0,759
9	0,0058	1,783	0,635	11,969	0,555
10	0,0020	0,611	0,732	14,116	0,634
11	0,0032	0,629	0,865	10,377	0,484
12	0,0054	0,774	1,372	12,006	0,556

* Portfolio 1 consists of the firms with low dividend yield while portfolio 12 consists of the firms with high dividend yield.

Table 7. Summary statistics for size-sorted portfolios

Portfolio number*	$\hat{\alpha}_i$	$t(\hat{\alpha}_i)$	$\hat{\beta}_i$	$t(\hat{\beta}_i)$	R_i^2
1	0,0149	1,989	0,622	5,119	0,186
2	0,0161	2,012	0,678	5,209	0,191
3	0,0164	1,166	0,788	3,461	0,094
4	0,0056	1,335	0,668	9,802	0,455
5	0,0063	1,413	0,662	9,173	0,423
6	0,0035	0,851	0,692	10,427	0,486
7	0,0069	1,366	0,942	11,459	0,533
8	0,0088	2,569	1,091	19,722	0,772
9	0,0048	1,357	0,859	14,988	0,661
10	0,0029	0,784	1,031	16,921	0,713
11	0,0026	0,534	1,257	16,159	0,694
12	0,0024	1,011	1,064	27,732	0,870

* Portfolio 1 consists of the small firms, while portfolio 12 consists of the large firms. $t(\hat{\alpha}_i)$ and $t(\hat{\beta}_i)$ are the t-statistics of the $\hat{\alpha}_i$ and $\hat{\beta}_i$ respectively. R_i^2 is the coefficient of determination.

Table 8. Summary statistics for liquidity-sorted portfolios

Portfolio number*	$\hat{\alpha}_i$	$t(\hat{\alpha}_i)$	$\hat{\beta}_i$	$t(\hat{\beta}_i)$	R_i^2
1	0,0049	1,80	0,59	13,18	0,60
2	0,0058	2,37	0,62	15,52	0,68
3	0,0101	1,83	0,67	7,54	0,33
4	0,0049	1,63	0,71	14,54	0,65
5	0,0055	1,24	0,66	9,23	0,43
6	0,0021	0,71	0,70	14,26	0,63
7	0,0047	1,58	0,66	13,71	0,62
8	0,0019	0,78	0,71	17,71	0,73
9	0,0065	2,06	0,78	15,23	0,67
10	0,0065	2,13	0,87	17,79	0,73
11	0,0065	1,85	0,79	13,87	0,63
12	0,0139	2,49	0,87	9,63	0,45

* Portfolio 1 consists of highly liquid firms while portfolio 12 consists of the most illiquid firms. $t(\hat{\alpha}_i)$ and $t(\hat{\beta}_i)$ of the $\hat{\alpha}_i$ and $\hat{\beta}_i$ respectively.

R_i^2 is the coefficient of determination.

larger companies. Consequently smaller companies are likely to have their betas underestimated to a larger extent. In addition the $\hat{\alpha}_i$ are found to be consistently positive in tables 6, 7 and 8. The univariate test that the α_i are significantly different from zero reveals that 5, 3 and 7 out of the 12 portfolios for the dividend-yield-sorted, size-sorted and liquidity-sorted data sets respectively, have significant alphas at the 5% level. To determine whether this evidence is significant in the multivariate setting, the multivariate test statistic was computed using each of the above-mentioned data sets.

Table 9 gives the summary statistics of the multivariate test for each of the component-sorted data sets. The results of the beta-sorted data set are also included for comparative purposes. Inspection of table 9 shows that although the smallest p -value is 0.12 for the dividend-yield-sorted data set, none of the test statistics are significant. Furthermore only the p -value of the dividend-yield-sorted data set is smaller than that of the beta-sorted data set, i.e. 0.325. This suggests that grouping shares either by dividend yield, size or liquidity does not have sufficient additional influence on returns to constitute a rejection of the *ex ante* efficiency of the JSE-Actuaries Overall index, and consequently the CAPM. These results are consistent with those of Bradfield, Affleck-Graves and Barr (1988).

Table 9. Summary of multivariate test results for the period 1976-1984

	beta-sorted	dividend-yield-sorted	size-sorted	liquidity-sorted
$\hat{\theta}^*$	0,373	0,428	0,356	0,349
W	0,131	0,175	0,119	0,114
$\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}$	0,132	0,176	0,120	0,115
Γ_1	1,155	1,545	1,051	1,006
p -value	0,325	0,120	0,409	0,449

Figure 2 shows the geometrical summary of the results for the various component-sorted data sets. The figure indicates that sorting the portfolios by either beta, dividend yield, size or liquidity does not have a significant impact on the relative difference in their slopes. Whether or not these slopes are significantly larger than the slope joining portfolio M depends largely on the power of the test. In this case the test result indicates that the slopes, in particular the distances along the slopes to given point, are not significantly different.

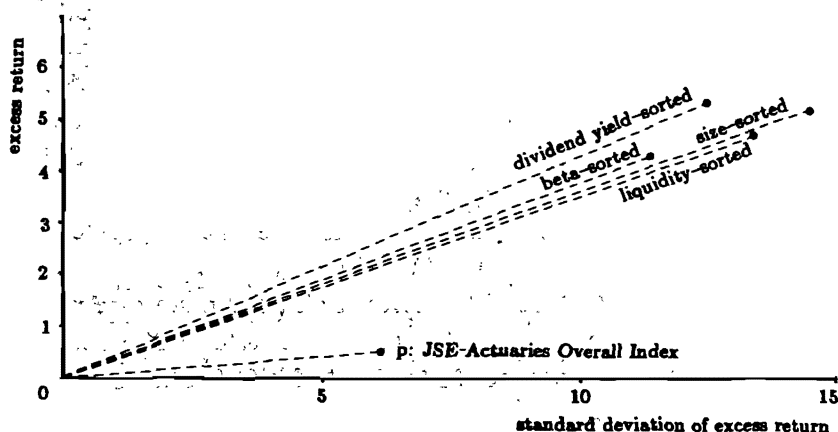


Figure 2. Geometrical summary over the period 1976-1984

6. CONCLUSIONS

The results of this multivariate analysis suggest that if the JSE-Actuaries Overall Index is used as a surrogate for the market portfolio, the CAPM appears to be a reasonable model in the South African context. This implies

that investors can expect to be compensated for bearing additional market risk (non-diversifiable risk) in the longer term.

The testing procedure also focused on several additional pricing factors, namely dividend yield, firm size and liquidity. None of the data sets sorted according to the above-mentioned components resulted in the rejection of the *ex ante* efficiency of the JSE-Actuaries Overall Index. It should, however, be noted that it has not been confirmed whether the absence of a liquidity effect or whether the possible inaccuracy of the estimate of liquidity is responsible for the lack of significance of the liquidity-sorted data set. Consequently it is concluded that none of these effects have a significant bearing on return in the CAPM framework from the South African perspective. Firstly, the absence of a dividend-yield effect implies that investors are not expected to enjoy a pre-tax gain by investing in high dividend yielding over low dividend yielding shares, other things being equal. Secondly, the absence of a size effect implies investors are not expected to earn abnormal returns by tilting their portfolios towards smaller firms, holding risk constant. Lastly, the absence of a liquidity effect implies that investors should not expect to gain higher returns by investing in less liquid stocks.

It is important that all of the above results, and hence conclusions, including those for the additional effects are consistent with the univariate test results and conclusions documented by Bradfield, Affleck-Graves and Barr (1988).

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